PROJECT DESIGN PHASE-II

Technology Stack (Architecture & Stack)

Date	25-09-2026
Team ID	
Project Name	Software Installation Request Automation (ServiceNow)
Maximum Marks	4 Marks

Technical Architecture

The architecture below presents the proposed components and data flow for the Software Installation Request Automation solution on ServiceNow. Employees submit software requests through a catalog item; request details are validated and stored, approvals are routed through workflow automation, and IT fulfillment teams update the request through completion.

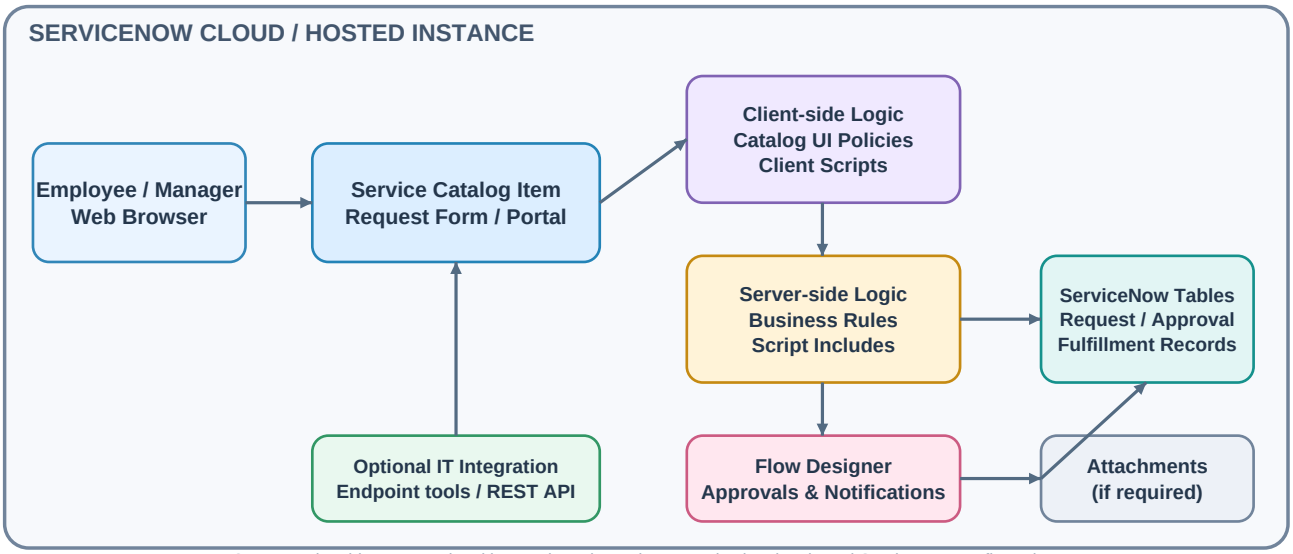

Figure 1. Conceptual technical architecture of the Software Installation Request Automation solution.

Table-1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Employees and approvers access the software catalog, request form, approval tasks, and status information through browser-based ServiceNow pages.	ServiceNow Service Catalog / Employee Center or Service Portal; HTML, CSS, JavaScript
2	Application Logic-1	Displays available software and captures requested application, business justification, device or user details, and required information.	Catalog Item, Catalog Variables, UI Policies, Client Scripts
3	Application Logic-2	Validates submitted values, applies request rules, and performs server-side processing on request records.	Business Rules, Script Includes, GlideRecord (JavaScript)

S.No	Component	Description	Technology
4	Application Logic-3	Routes requests for approval, handles approval outcomes, sends notifications, and coordinates fulfillment tasks.	Flow Designer, Approval actions, ServiceNow Notifications
5	Database	Stores software request details, approval records, fulfillment tasks, status history, and related configuration data.	ServiceNow platform tables / records (for example, request and task records)
6	Cloud Database	Platform-managed storage for application records within the ServiceNow instance; a separate database is not required by the described scope.	ServiceNow instance data storage
7	File Storage	Stores optional supporting documents or installation-related attachments when required and permitted by access rules.	ServiceNow Attachment storage
8	External API-1	Email notifications communicate submission, approval decisions, and fulfillment or completion updates where email is configured.	ServiceNow Email / SMTP configuration
9	External API-2	Optional integration with endpoint or software deployment tools may support automated installation if the organization enables and configures it.	Approved integration / REST API (optional; e.g. endpoint-management platform)
10	Machine Learning Model	The current project scope focuses on request, approval, and fulfillment workflow automation; a separate machine-learning model is not required.	Not applicable to current project scope
11	Infrastructure (Server / Cloud)	The catalog, workflows, records, and server-side logic run on the hosted ServiceNow platform; users connect through a web browser.	ServiceNow cloud instance; browser-based client

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	The solution primarily uses the ServiceNow platform and its catalog / portal framework. HTML, CSS, and JavaScript may be used for interface customization; no additional open-source framework is specified.	ServiceNow Service Catalog / Employee Center or Service Portal, HTML, CSS, JavaScript
2	Security Implementations	Access should be controlled through platform authentication, roles, and ACLs. Employees should view their own requests, while approvers and IT fulfillment staff should access records according to their responsibilities.	ServiceNow roles, Access Control Lists (ACLs), platform authentication
3	Scalable Architecture	The solution separates catalog presentation, request processing, approval workflows, and fulfillment records using ServiceNow platform components. Capacity depends on instance configuration and organizational usage.	ServiceNow platform architecture; modular catalog items and Flow Designer
4	Availability	The application is accessed through the hosted ServiceNow instance, subject to platform service availability and scheduled maintenance.	ServiceNow cloud hosting
5	Performance	Use efficient server-side queries, avoid unnecessary client/server calls, and keep catalog forms and request views focused on required information. Actual performance depends on instance configuration and load.	GlideRecord query practices, optimized scripts and catalog components